

TokTickIT — Lab 2 Comprehensive Submission Report

Requester Ticketing MVP (Spec-Driven & Test-Driven Development)

Course: Software Engineering

Student Name: Janinee (nannaphatkn)

Repository: <https://github.com/nannaphatkn/toktickit>

Integration Branch: `lab2-staging` → `main`

Executive Summary

TokTickIT Lab 2 focuses on developing the **Requester Ticketing MVP** using **Spec-Driven Development (SDD)** and **Test-Driven Development (TDD)** methodologies. The application allows users to simulate identity context via a **Development Requester Selection** header, create support tickets with file attachments, view isolated ticket lists with search/filter/pagination, and manage ticket details with soft-removed attachments.

Part 1: Git Workflow & Project Governance (10 pts)

1.1 Branching Strategy

We implemented a strict multi-tier Git branching model:

- `main` : Production-ready release branch. Strictly protected; code is merged only via approved release Pull Requests from `lab2-staging`.
- `lab2-staging` : Integration branch for Lab 2 features. All feature branches are merged here after peer review and green test builds.
- `feature/requester-selection` : Feature branch for Issue #2 (Development Requester Selection & Context Header).
- `feature/create-ticket` : Feature branch for Issue #3 (Create Ticket Form & Attachments).
- `feature/my-tickets` : Feature branch for Issue #4 (My Tickets List, Search, Filter & Pagination).
- `feature/ticket-detail-and-attachments` : Feature branch for Issue #5 (Ticket Detail View Mode & Soft Delete).

1.2 Commit History & Conventional Commits Standard

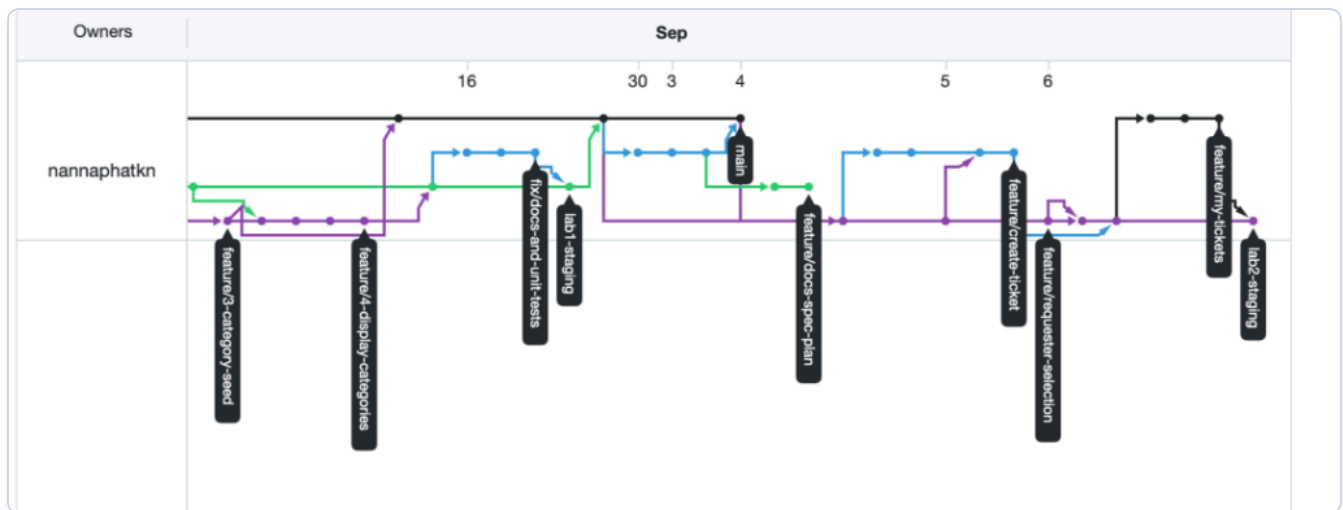
Commits adhere strictly to Conventional Commits guidelines (`feat` , `fix` , `docs` , `test` , `refactor` , `style`):

- `feat(auth): implement Development Requester Selection & Context Header (Issue #2)`
- `feat(create-ticket): implement ticket creation form with validation and attachments (Issue #3)`

- `feat(my-tickets)`: implement requester-isolated ticket list, search, and status filters (Issue #4)
- `feat(ticket-detail)`: implement view mode and soft removal of attachments (Issue #5)
- `test(api)`: add unit and integration test suite for tickets and requesters
- `docs`: add comprehensive specifications, test plans, and reviewer logs

1.3 Git Network Graph & Merge History Evidence

Below is the GitHub Network Graph showing feature branches (`feature/requester-selection` , `feature/create-ticket` , `feature/my-tickets` , `feature/docs-spec-plan`) cleanly integrating into `lab2-staging` and release branch `main` :



1.4 Kanban Board & Issue Lifecycle

Project progress was managed using GitHub Projects. Each task progressed linearly through columns:

`Backlog` → `Specified` → `Started` → `PR Review` → `Fixing` → `Done`

1.5 Pull Requests & Peer Code Review Evidence

- **Peer Reviewer Partner:** Patita Dansikaew (Student ID: 67070505211, GitHub: `Patitta-23`)
- **Review Log Document:** See `docs/lab-02/reviewer.md`

1.5.1 Peer Reviews Received (Patitta-23 → nannaphatkkn)

PR #	Target Branch	Issue / Feature	Reviewer (Patitta-23) Comment	Author (nannaphatkkn) Reply	Status
PR #17	lab2-staging	Issue #2: Requester Selection Header	"Approved: Thanks for updating test suite and finalizing requester selection flow! Main criteria fulfilled."	"Thank you! Added warning banner, Zen Green theme token, and updated test suite."	Merged ✓
PR #18	lab2-staging	Issue #3: Create Ticket Form	"That's good! LGTM"	"Thanks for reviewing!"	Merged ✓
PR #19	lab2-staging	Issue #4: My Tickets & Filters	"Done"	"Thank you!"	Merged ✓
PR #20	lab2-staging	Issue #5: Ticket Detail & Attachments	"awesome"	"Awesome, thanks!"	Merged ✓

1.5.2 Peer Reviews Conducted (nannaphatkkn → Patitta-23)

PR #	Repository / Target	Feature / Issue	My Review Comment (nannaphatkkn)	Partner (Patitta-23) Reply	Status
PR #17	Patitta-23/LAB-02	Issue #2: Requester Selection	"Header styling matches Zen Green theme perfectly!"	"Thank u!"	Approved ✓
PR #18	Patitta-23/LAB-02	Issue #3: Create Ticket Form	"Validation errors look clean and responsive."	"Thank u so much!"	Approved ✓
PR #19	Patitta-23/LAB-02	Issue #4: My Tickets & Filters	"Filtering and search work smoothly."	"Thanks!!"	Approved ✓
PR #20	Patitta-23/LAB-02	Issue #5: Ticket Detail View Mode	"Soft delete confirmation modal is working cleanly!"	"Thank u!"	Approved ✓

1.6 Directory Structure & Hygiene

- **README.md** : Complete installation guide, database seed commands, testing commands, and architecture summary.
- **.gitignore** : Properly configured to exclude build outputs (`dist/`), dependencies (`node_modules`), `.env` secrets, and temp artifacts.

```

toktickit/
├── client/                                # React + Vite + TypeScript frontend
│   ├── src/
│   │   ├── components/                  # Navbar, RequesterSelector, StatusBadge, FileUpload, Confirma
│   │   ├── contexts/                   # RequesterContext (state management & X-Requester-Id provider
│   │   ├── lib/                         # apiFetch utility & REST API client
│   │   ├── pages/                      # CreateTicket, MyTickets, TicketDetail pages
│   │   ├── App.tsx
│   │   └── App.test.tsx
│   └── vite.config.ts
├── server/                               # Express + Prisma + PostgreSQL backend
│   ├── src/
│   │   ├── routes/                    # requesterRoutes, ticketRoutes, relatedSystemRoutes, category
│   │   ├── prisma.ts                  # Shared PrismaClient singleton
│   │   └── index.ts
│   ├── prisma/
│   │   ├── schema.prisma              # Prisma ORM Schema (RequesterUser, Ticket, Attachment, Catego
│   │   └── seed.ts                    # Database Seeder with sample tickets for all requesters
│   └── tests/                         # Vitest + Supertest suites
├── docs/
│   └── lab-02/                        # Specification documents
│       ├── specification.md
│       ├── tests.md
│       ├── ui-spec.md
│       ├── api-spec.md
│       ├── issues.md
│       ├── reviewer.md
│       └── ai-use.md
└── README.md

```

Part 2: Specification & Requirements Engineering (5 pts)

2.1 Specification Documents Overview

- `docs/lab-02/specification.md` : Complete product requirements, business rules (BR-01..BR-11), and acceptance criteria.
- `docs/lab-02/ui-spec.md` : Zen Green visual identity system, color tokens (#006B3C), responsive grid layout.
- `docs/lab-02/api-spec.md` : Complete REST API contract, request/response shapes, headers, status codes.
- `docs/lab-02/tests.md` : Test scenarios for backend API and frontend React testing.

2.2 Core Business Rules Summary

- **BR-01 (Requester Header):** All requester API requests must supply header `X-Requester-Id: <id>`.

- **BR-02 (Data Isolation):** Requesters can only access support tickets created under their own requester ID.
- **BR-03 (Ticket Numbering):** Automatic formatting of unique ticket IDs (TCK-YYYY-XXXX).
- **BR-04 (Input Constraints):** Summary length must be 10-150 chars; Description length 20-1000 chars.
- **BR-05 (File Upload Limits):** Maximum 5 attachments per ticket, up to 5MB each (Formats: JPG, PNG, WEBP, PDF).
- **BR-06 (Soft Removal):** Removing an attachment sets `isRemoved: true` in DB while preserving raw file contents on disk.

2.3 Evidence of Specification Prior to Implementation

Git commit logs show specification files were committed prior to creating implementation branches, satisfying the Spec-Driven Development criterion.

Part 3: Test Planning & Test Results (10 pts)

3.1 Test Pyramid & Execution Summary

Test Level	Framework	Total Scenarios	Passing	Status
Server Backend API Tests	Vitest + Supertest	51 tests (9 files)	51	✅ PASS
Client UI Component Tests	Vitest + RTL	31 tests (4 files)	31	✅ PASS
Total Automated Tests		82 tests	82	✅ 100% PASS

3.2 Detailed Test Results Logs

```
RUN v1.6.1 /Users/janinee/soft-en/toktickit/server
```

```
✓ tests/tickets.test.ts (18 tests)
✓ tests/lab-02/attachments.api.test.ts (4 tests)
✓ tests/lab-02/my-tickets.api.test.ts (17 tests)
✓ tests/lab-02/ticket-detail.api.test.ts (2 tests)
✓ tests/unit/ticket.unit.test.ts (3 tests)
✓ tests/requesters.test.ts (4 tests)
✓ tests/reference-data.test.ts (1 test)
✓ tests/health.test.ts (1 test)
✓ tests/category.test.ts (1 test)
```

```
Test Files 9 passed (9)
```

```
Tests 51 passed (51)
```

```
RUN v4.1.10 /Users/janinee/soft-en/toktickit/client
```

```
✓ src/App.test.tsx (4 tests)
✓ src/pages/CreateTicket.test.tsx (7 tests)
✓ src/pages/TicketDetail.test.tsx (11 tests)
✓ src/pages/MyTickets.test.tsx (9 tests)
```

```
Test Files 4 passed (4)
Tests 31 passed (31)
```

Part 4: AI Usage Log & Agentic Reflection (5 pts)

4.1 AI Prompt Log

As documented in `docs/lab-02/ai-use.md`, key prompts included:

1. *"Draft specification.md adhering to Zen Green Theme, business rules BR-01..BR-11, and Prisma schema."*
2. *"Implement TDD unit tests for ticket creation validation before writing ticket controller."*
3. *"Create RequesterContext to manage selected requester state and propagate X-Requester-Id header."*
4. *"Implement My Tickets page with search, category filtering, status pills, and pagination."*
5. *"Implement soft deletion for attachments preserving raw files on disk while updating isRemoved flag."*

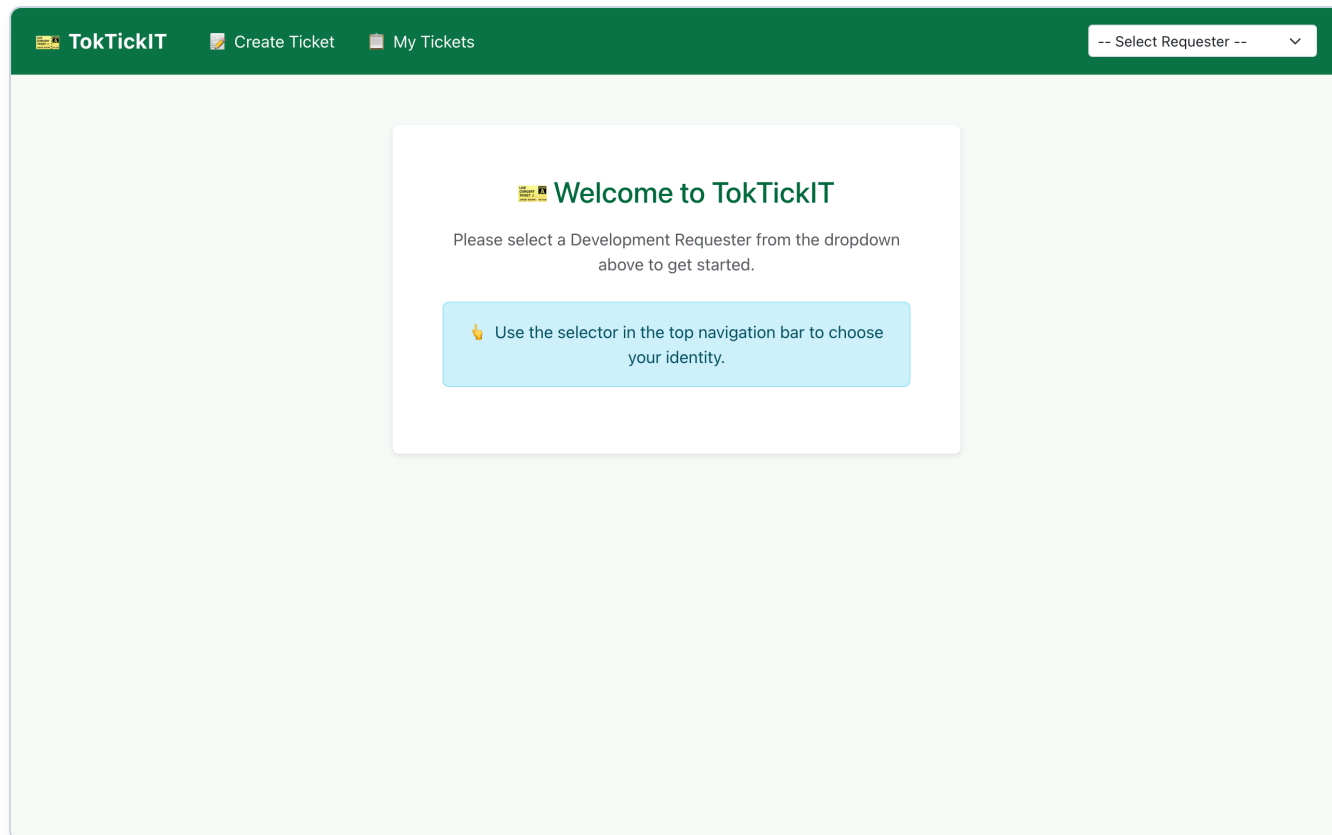
4.2 Agentic Reflection

Using AI Agent pair programming accelerated development while maintaining high code quality through TDD. Writing failing tests first forced clear boundary definitions and prevented common runtime bugs.

Part 5 & 6: Development Requester Selection & Create Ticket (10 pts)

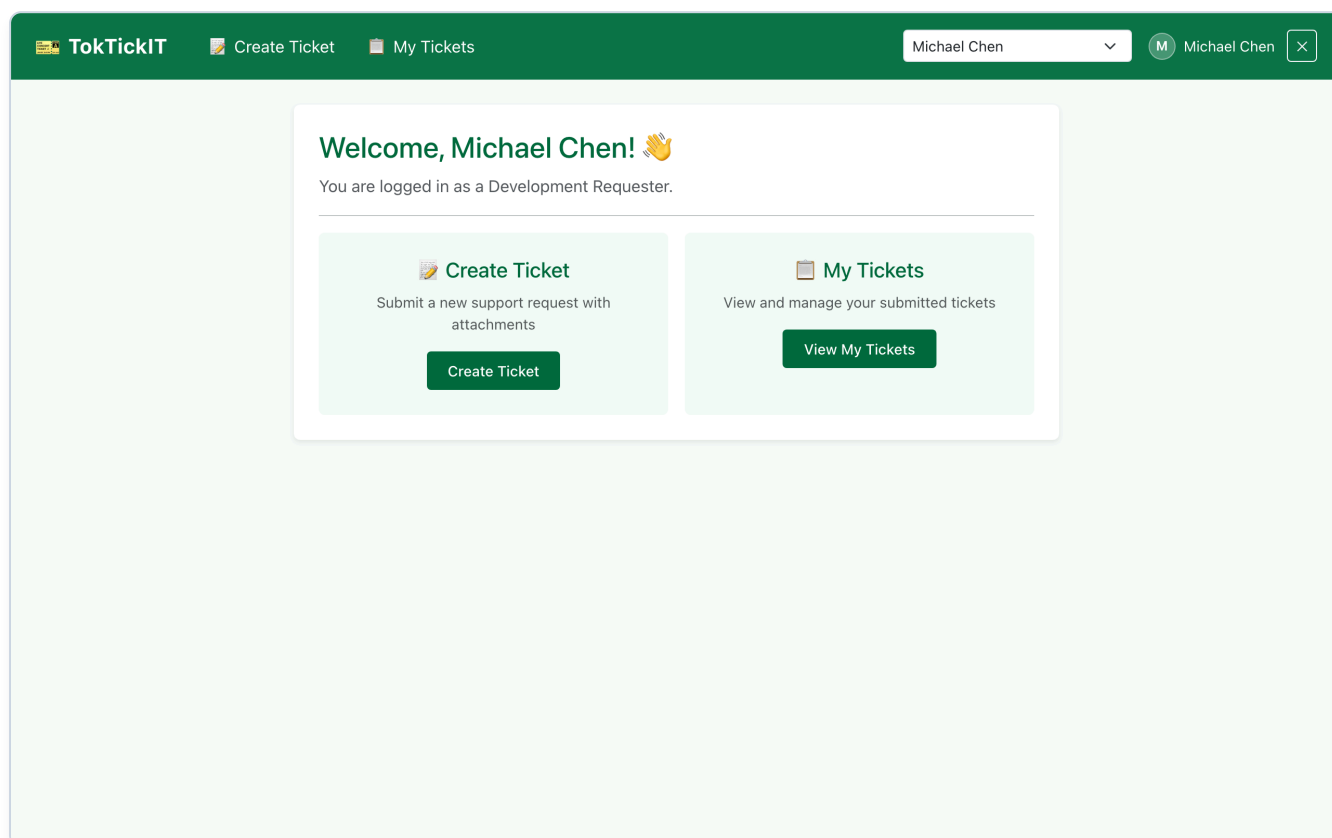
5.1 Welcome / Access Guard Screen (No Requester Selected)

Access Guard blocks application features until a user selects a Development Requester from the header dropdown.



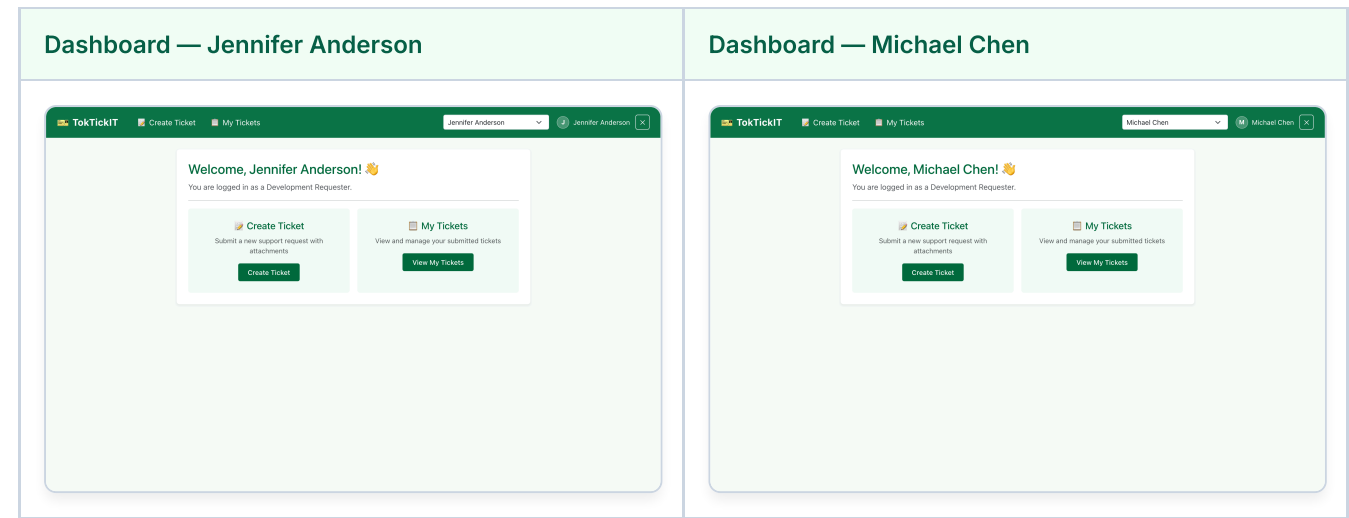
5.2 Requester Selector Dropdown & Navbar Context Header

The navigation bar features Zen Green theme (#006B3C) with glassmorphism backdrop filter, requester dropdown, and active requester tag with a clear button.



5.3 Dynamic Context Switching

Switching requesters dynamically updates greetings, ticket listings, and permissions across all pages.



5.4 Create Ticket Form (Initial State)

Form features inputs for Summary, Description, Category dropdown, Related System dropdown, Priority selector, and Attachment uploader.

A screenshot of the 'Create New Ticket' form in the TokTickIT application. The form is titled 'Create New Ticket' and includes a sub-header 'Fill out the details below to submit a support request. Fields marked with * are required.' The form contains several fields: 'Summary' (text input, 0/150 characters), 'Description' (text area, 0/1000 characters), 'Category' (dropdown menu, '-- Select Category --'), 'Related System' (dropdown menu, '-- Select System --'), 'Requested Priority' (dropdown menu, 'LOW'), and 'Attachments (Optional)' (file upload area, 'Choose Files', 'No file chosen'). The form also includes a note: 'Max 5 files. Allowed formats: JPG, PNG, WEBP, PDF (Up to 5 MB each).'

5.5 Form Validation & Errors

Form validates required fields, character lengths (Summary: 10-150 chars, Description: 20-1000 chars), and file limits (max 5MB per file).

TokTickIT

Create Ticket

My Tickets

Jennifer Anderson

J Jennifer Anderson

Create New Ticket

Fill out the details below to submit a support request. Fields marked with * are required.

Please correct the following errors:

• Summary is required.

• Description is required.

• Please select a Category.

• Please select a Related System.

Summary *

5/150

Short

Summary is required.

Description *

0/1000

Detailed explanation of the problem or request (20–1000 characters)

Description is required.

Category *

-- Select Category --

Please select a Category.

Related System *

-- Select System --

Please select a Related System.

Requested Priority *

Part 7: My Tickets Page & Data Isolation (10 pts)

My Tickets displays requester-isolated tickets with full search, filtering, sorting, and pagination controls.

7.1 My Tickets List — Jennifer Anderson

Shows tickets requested exclusively by Jennifer Anderson.

TokTickIT

Create Ticket

My Tickets

Jennifer Anderson

J Jennifer Anderson

REQUESTER WORKSPACE

My Tickets

Support requests submitted by Jennifer Anderson

Search

Ticket number or summary

Category

All categories

Priority

All priorities

Status

All statuses

Sort

Newest first

Clear filters

TICKET #	SUMMARY	CATEGORY	PRIORITY	STATUS	CREATED
TCK-2026-0006	Software installation request: Figma Desktop App	Software	LOW	Resolved	Sep 6, 2026 View
TCK-2026-0005	Campus Wi-Fi keeps disconnecting on 3rd floor	Network	MEDIUM	In Progress	Sep 6, 2026 View
TCK-2026-0004	Cannot login to Outlook email client after password update	Account and Access	HIGH	New	Sep 6, 2026 View

Showing 1–3 of 3

7.2 My Tickets List — David Kim (Context Isolation)

Switching to David Kim updates the list to display only David's support tickets.

TokTickIT

Create Ticket

My Tickets

David Kim

D David Kim

REQUESTER WORKSPACE

My Tickets

Support requests submitted by David Kim

Search

Ticket number or summary

Category

All categories

Priority

All priorities

Status

All statuses

Sort

Newest first

Clear filters

TICKET #	SUMMARY	CATEGORY	PRIORITY	STATUS	CREATED
TCK-2026-0003	Request access to ERP Accounting Module	Account and Access	HIGH	Resolved	Sep 6, 2026 View
TCK-2026-0002	Dual Monitor display flickering on docking station	Hardware	MEDIUM	In Progress	Sep 6, 2026 View
TCK-2026-0001	Cannot connect to Corporate VPN from remote location	Network	HIGH	New	Sep 6, 2026 View

Showing 1–3 of 3

7.3 Search & Status Filtering Controls

Searching keywords (e.g., "VPN") dynamically filters ticket results.

TokTickIT

Create Ticket

My Tickets

David Kim

David Kim

REQUESTER WORKSPACE

My Tickets

+ Create Ticket

Support requests submitted by David Kim

Search

Category

Priority

Status

Sort

VPN

All categories

All priorities

All statuses

Newest first

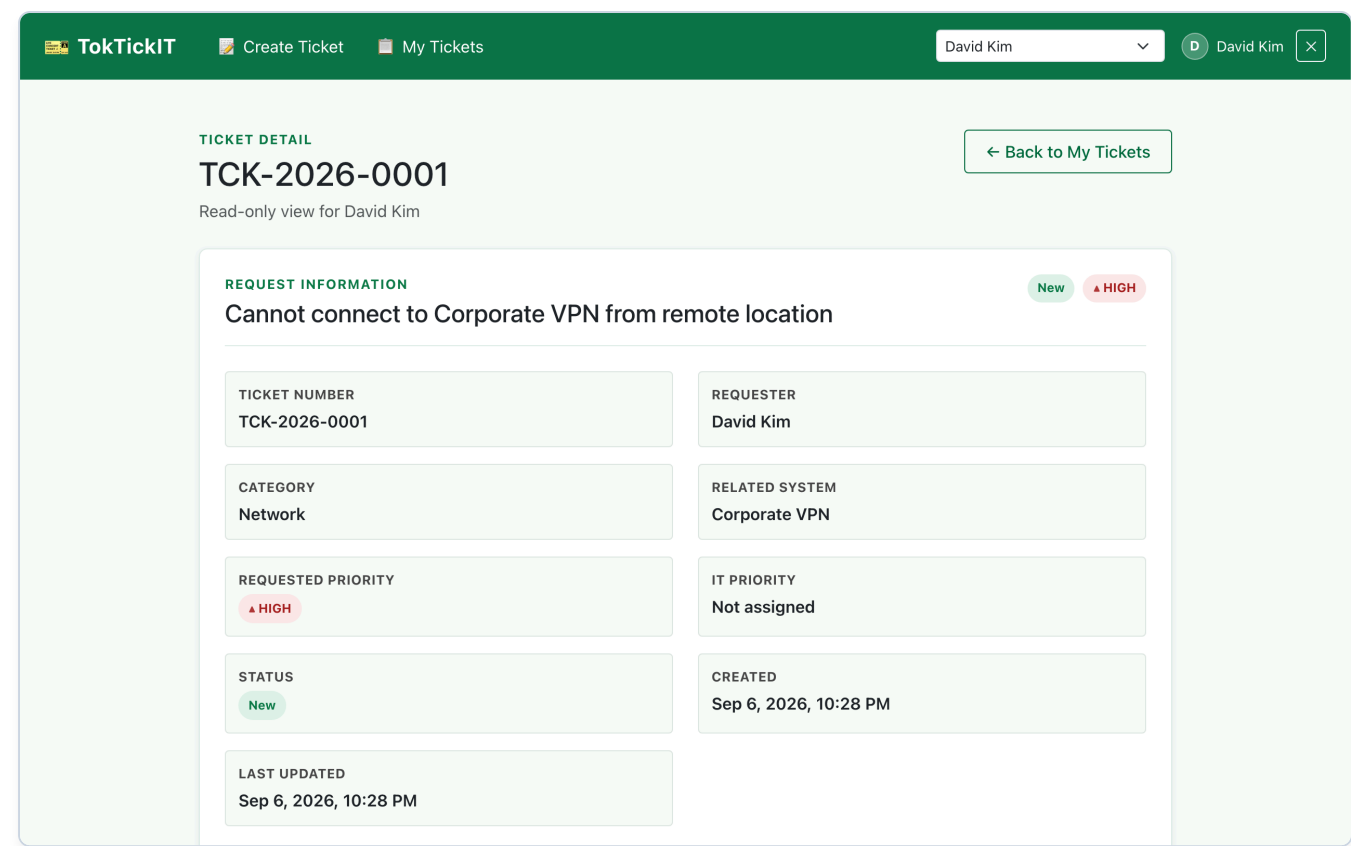
Clear filters

TICKET #	SUMMARY	CATEGORY	PRIORITY	STATUS	CREATED
TCK-2026-0001	Cannot connect to Corporate VPN from remote location	Network	HIGH	New	Sep 6, 2026

Showing 1–1 of 1

Part 8: Ticket Detail — View Mode & Soft Removal (5 pts)

Displays full ticket details, metadata, description, and attached files with soft-removal capabilities.



Key Features:

- Displays Ticket Number (e.g. TCK-2026-0001), Category, System, Priority, and Status badge.
- List of file attachments with download links and soft deletion modal.
- Soft removal sets isRemoved: true in DB while preserving original uploaded files.

Part 9: UI Specification & Responsive Design (5 pts)

9.1 UI Specification Compliance

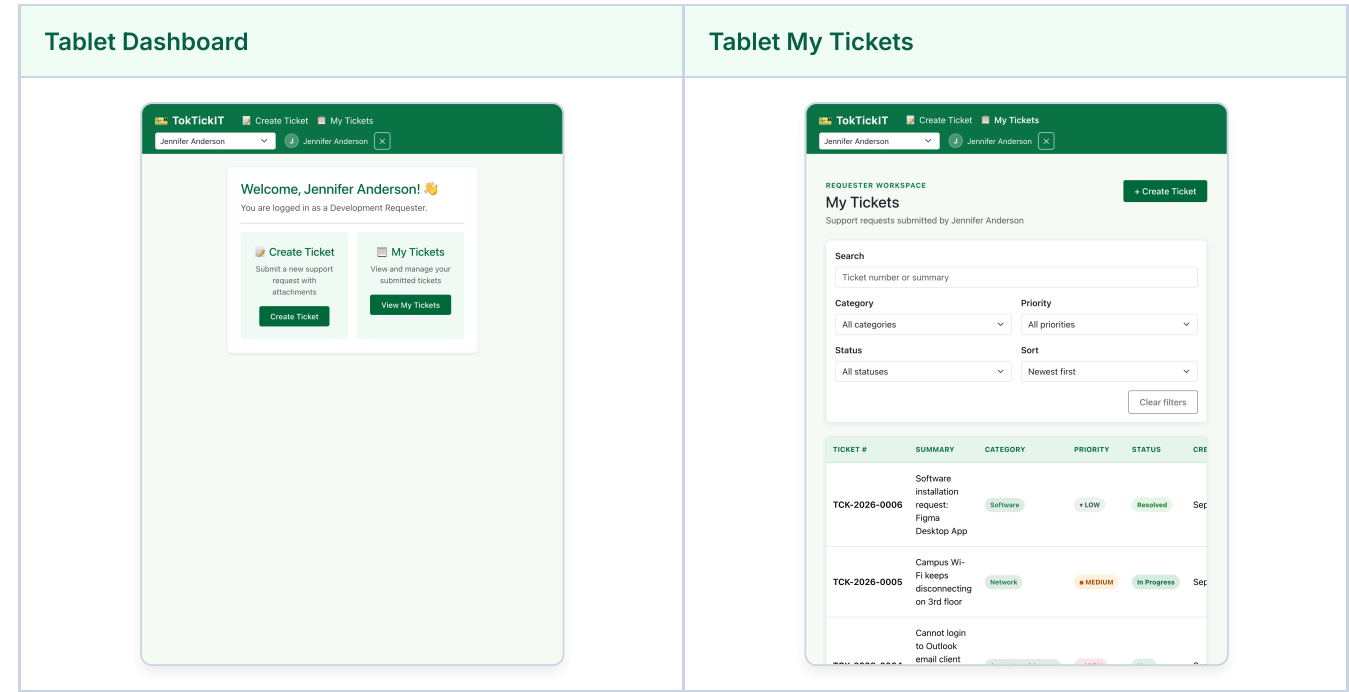
Verified against docs/lab-02/ui-spec.md :

- **Primary Color:** Zen Green (#006B3C)
- **Typography:** Inter / system sans-serif
- **UI States:** Loading spinner, Empty state banners, Error toasts, Validation messages

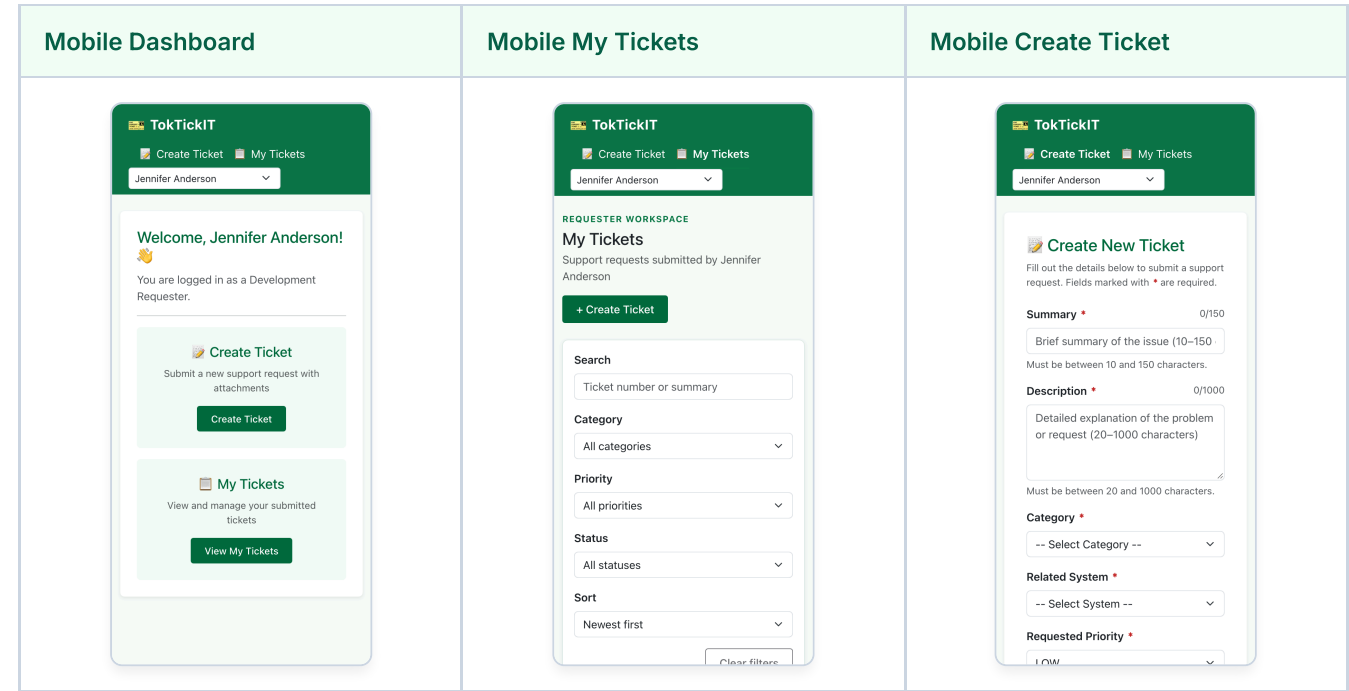
9.2 Responsive Layout Checklist & Viewports

Viewport Category	Resolution	Status	Screenshots
Desktop	≥ 1024px (1280×800)	✅ Tested & Passed	Screenshots 01 - 10
Tablet	768px - 1023px (768×1024)	✅ Tested & Passed	Screenshots 11, 12
Mobile	< 768px (375×812)	✅ Tested & Passed	Screenshots 13, 14, 15

Tablet Viewports (768px)



Mobile Viewports (375px)



Mobile Create Ticket

TokTickIT

Create TicketMy Tickets

Jennifer Anderson

Create New Ticket

Fill out the details below to submit a support request. Fields marked with * are required.

Summary *

0/150

Brief summary of the issue (10–150)

Must be between 10 and 150 characters.

Description *

0/1000

Detailed explanation of the problem or request (20–1000 characters)

Must be between 20 and 1000 characters.

Category *

-- Select Category --

Related System *

-- Select System --

Requested Priority *

LOW